

Network Health Assessment

Objective

A network health assessment was conducted on the Administrative Network segment (192.168.10.0/24) to identify active hosts, exposed services, and areas that may require further security review.

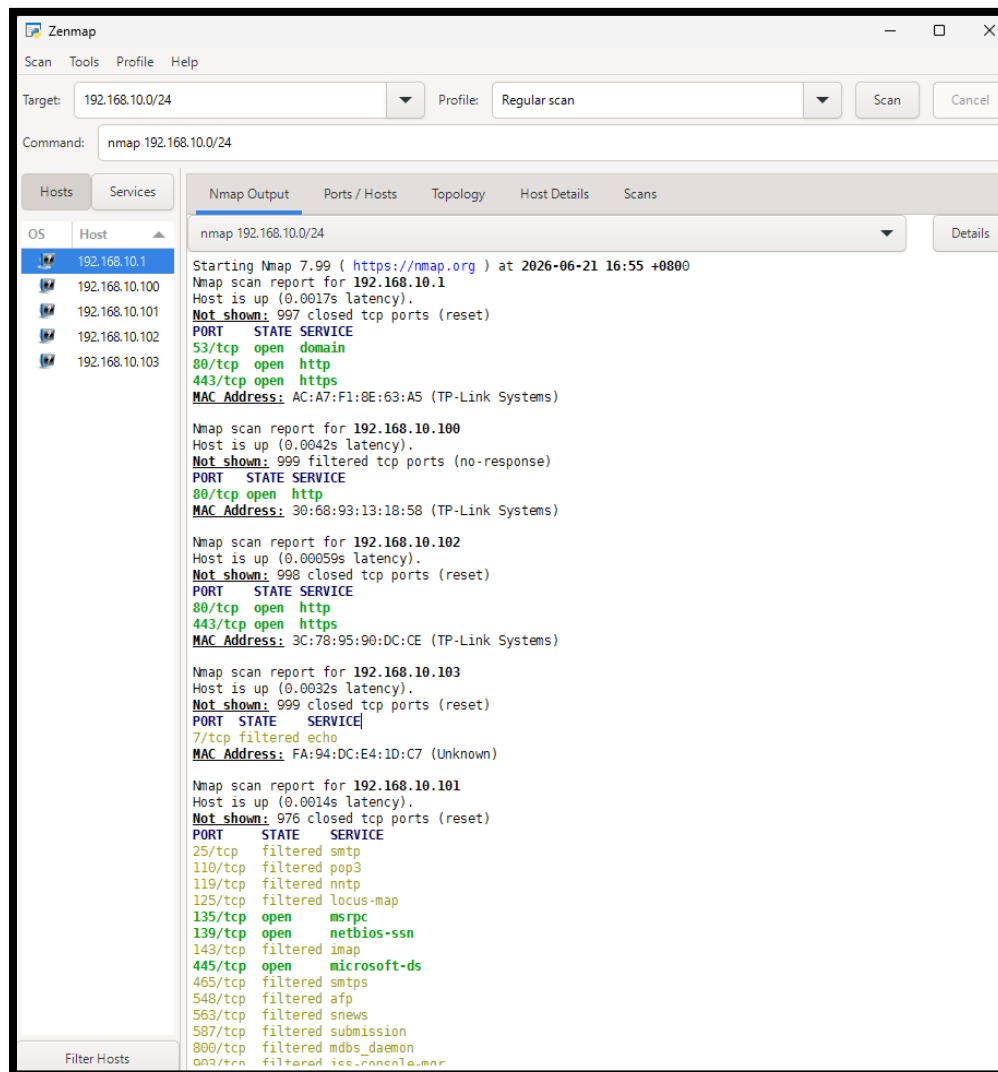
Network Overview

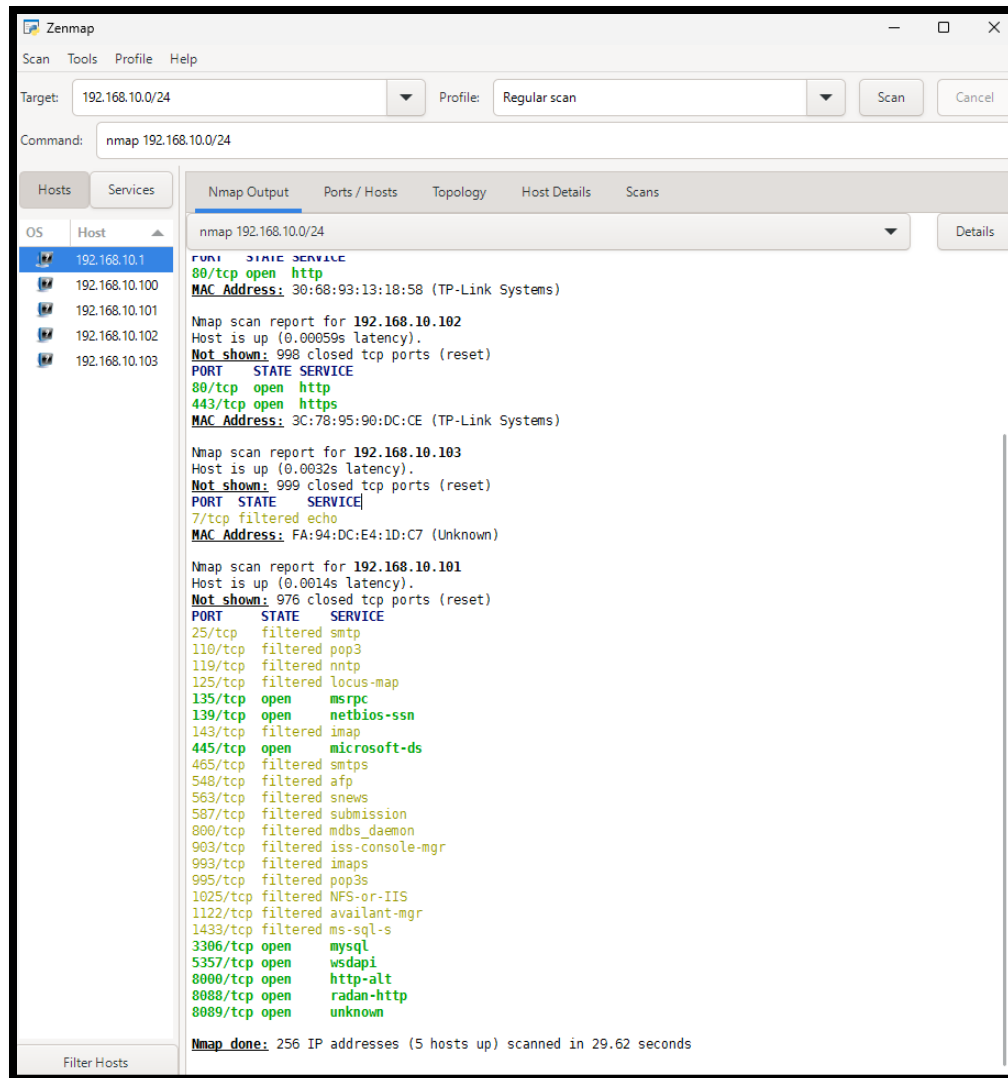
The administrative network is connected through the following infrastructure:

ISP Router → TP-Link Router → Managed Switch → Endpoints and Access Point

Methodology

A network discovery and service enumeration scan was performed using Nmap (regular scan) to identify active hosts and exposed TCP services within the subnet.





Findings

A. Host 1 – Network Gateway (192.168.10.1)

The host was identified as Omada TP-Link Router that serves as a gateway for the administrative subnet.

The following are the observed services:

- DNS (Port 53)
- HTTP (Port 80)
- HTTPS (Port 443)

Interpretation

The exposed services are consistent with the expected role of a network router. The DNS supports the name resolution while both HTTPS and HTTP provides web-based administration capabilities.

B. Host 2 – Managed Switch (192.168.10.100)

The host was identified as the TP-Link Managed Switch.

Below is the observed service:

- Port 80 (HTTP)

Interpretation

The managed switch exposes a web-based management interface while filtering the majority of other scanned ports. This behavior is typical and consistent with a managed network switch that limits access and service to administrative functions.

C. Host C. – Access Point (192.168.10.102)

The host was identified as a TP-Link Access Point that functions as a wireless access point device.

Observed Services:

- HTTP (Port 80/TCP)
- HTTPS (Port 443/TCP)

Interpretation:

The device exposes both HTTP and HTTPS management interfaces, which is typical of wireless access point or secondary network devices. The presence of dual management ports suggests that the configuration capability is managed by the administrator via a web interface.

D. Host F. – Mobile Phone (192.168.10.103)

The host presents limited visibility with minimal response to scanned ports. Through further checking, the host was immediately identified as the mobile phone connected to the network through the AP, which is Host C.

- TCP/7 (filtered)
- (Unknown MAC vendor) – from manufacturer.

Interpretation:

Although the device does not expose identifiable services within the scanned range and may initially and may appear as an IoT device, further investigation and monitoring resulted in identifying the host.

E. Host 5 – Laptop (Controller Endpoint) (192.168.10.101)

The host has windows operating system installed.

The following are observed by the following:

- Microsoft RPC (Port 135)
- NetBIOS Session Service (139)
- SMB / Microsoft-DS (Port 445)

- MySQL (Port 3306)
- WSDAPI (Port 5357)
- HTTP Alternative Service (Port 8000)
- Splunk Universal Forwarder (Port 8089)

Interpretation

The enumerated exposed services indicate that the workstation provides standard Windows networking functionality together with locally hosted application services installed in the host. The presence of MySQL and web-related services is present since the host was also used in web development activities that include hosting.

Overall Assessment

Five active hosts were identified within the administrative subnet.

- TP-Link Router (Gateway)
- TP-Link Managed Switch
- TP-Link Access Point
- Mobile Phone
- Windows Workstation (Laptop)

The network demonstrated a typical segmented home office environment with infrastructure devices and endpoint systems that coexist within the same subnet. While most infrastructure devices expose only management interfaces, the workstation presents several service footprints gathered from other activities executed in the device, aside from monitoring and administrative purposes.

The results suggest that there are no immediate critical misconfigurations during the initial scan. However, if other unidentified hosts appear on the next scan that are neither enumerated nor expected to be connected through the access point, deeper scanning is strongly recommended.